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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5070 CHEMISTRY

5070/31 Paper 3 (Practical Test), maximum raw mark 40

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1 (a) Titration

Accuracy 8 marks

For the two best tires give:
4 marks for a value within 0.2 cm^3 of supervisor
2 marks for a value within 0.3 cm^3 of supervisor
1 mark for a value within 0.4 cm^3 of supervisor

Concordance 3 marks

Give:
3 marks if all the ticked values are within 0.2 cm^3
2 marks if all the ticked values are within 0.3 cm^3
1 mark if all the ticked values are within 0.4 cm^3

Average 1 mark

Give 1 mark if the candidate calculates a correct average (error not greater than 0.05) of all his ticked values.

Assuming a 25 cm^3 pipette and a titre of 24.8 cm^3.

(b) concentration of sulfuric acid in **P** [2]

$= \frac{25 \times 0.1}{2 \times 24.8}$ (1)

= 0.0504 (1)

Answers should be correct to + or – 1 in the third significant figure.

(c) concentration of sulfuric acid in battery acid [1]

= 0.0504 × 100 (1)

= 5.04

answer from **(b)** × 100

(d) mass of sulfuric acid present in 4.50 dm^3 of battery acid [1]

= 5.04 × 4.5 × 98 (1)

= 2220

answer from **(c)** × 4.5 × 98

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2 **R** is copper(II) sulfate **S** is copper(I) oxide

Test	Notes
General points For ppt allow solid, suspension, powder For gases Name of gas requires test to be at least partially correct. Effervesces = bubbles = gas vigorously evolved but not gas evolved Solutions Colourless not equivalent to clear, clear not equivalent to colourless	
Solution **R**	
Test 1 **(a)** white ppt (1) **(b)** insoluble in nitric acid (1)	
Test 2 blue ppt (1) soluble in excess (1) dark blue solution (1)	
Test 3 **(a)** solution turns green (1) **(b)** blue ppt (1) insoluble in excess (1)	allow green-blue or green-yellow
Test 4 solid turns red or brown (1) blue colour fades (1)	
Test 5 solid turns brown (1) blue solution (1)	allow colour darkens
Test 6 solid turns brown (1) blue solution (1) solid disappears (1) effervescence (1) yellow or brown gas (1)	allow colour darkens

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Test 7 **(a)** solid turns white or off-white (1) **(b)** solid disappears (1) blue solution (1)	
Test 8 **(a)** no reaction (1) **(b)** effervescence (1) gas relights a glowing splint (1) oxygen (1) blue solution (1)	

Conclusions

Anion in **R** is sulfate or SO_4^{2-} (in **test 1** ppt in **(a)** must not dissolve in acid) (1)
The metal in **R** and **S** is copper, copper(II), Cu or Cu^{2+} (1)

Note: 27 marking points, maximum 24.